

**Assignment 4**  
**Due at 11:59 pm on Wednesday, October 7**

**Reading:** Chapters 6.1–6.3 of the textbook

1. (18 points) This problem has three parts

**Part A** (6 points, from Problem 3 in section 6.11 of the textbook): Find all pure strategy Nash equilibria in the game below. In the payoff matrix below the rows correspond to player A's strategies and the columns correspond to player B's. The first entry in each box is player A's payoff and the second entry is player B's payoff.

|          |   | Player B |      |
|----------|---|----------|------|
|          |   | L        | R    |
| Player A | U | 1, 2     | 3, 2 |
|          | D | 2, 4     | 0, 2 |

**Part B** (6 points, from Problem 4 in section 6.11 of the textbook): Consider the two-player game with players, strategies and payoffs described in the following game matrix.

- Does either player have a dominant strategy? Explain briefly (1-3 sentences).
- Find all pure strategy Nash equilibria for this game.

|          |   | Player B |      |      |
|----------|---|----------|------|------|
|          |   | L        | M    | R    |
| Player A | t | 0, 3     | 6, 2 | 1, 1 |
|          | m | 2, 3     | 0, 1 | 7, 0 |
|          | b | 5, 3     | 4, 2 | 3, 1 |

**Part C** (6 points, from Problem 5 in section 6.11 of the textbook): Consider the following two-player game in which each player has three strategies. Find all the pure strategy Nash equilibria for this game.

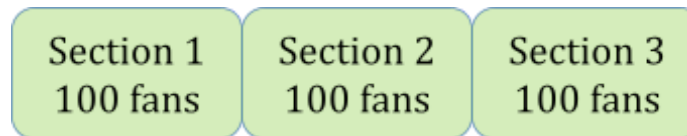
|          |   | Player B |      |      |
|----------|---|----------|------|------|
|          |   | L        | M    | R    |
| Player A | U | 1, 1     | 2, 3 | 1, 6 |
|          | M | 3, 4     | 5, 5 | 2, 2 |
|          | D | 1, 10    | 4, 7 | 0, 4 |

2. (20 Points) Consider a football stadium that has 3 seating sections (S1, S2, S3) in a row. The distance between section S1 and S2 is the same as the distance between sections S2 and S3. Each of the sections seats 100 people.

Now, consider two hotdog vendors who have to decide the sections in which to set up their shop in order to sell as many hotdogs as possible. Since the fans do not want to miss any part of the game, they will buy from the hotdog vendor in the sections closest to them (assume the distance inside the section does not matter to them). We will also assume that this is a hungry crowd in a popular game, so every seat is occupied and every person will buy one hotdog.

If both the vendors are in equally distant sections, then half the fans buy from one vendor and half from the other. For example, if one vendor is in S1 and one vendor is in S3, half the people in S2 will buy from the first vendor and half from the second vendor. Similarly, if both vendors are in the same section, half of the fans will buy one vendor and half from the other vendor.

- Construct a payoff matrix showing the payoffs of the vendors for each of their strategies. Underline all best responses.
- Is there a dominant strategy for this game? If so, is it strictly dominant? Explain.
- Find the pure-strategy Nash Equilibria of this game. Explain.



3. (25 points) For each of the following, either (i) give an example of the payoff matrix of a game with **2 players and 3 strategies per player** that satisfies the requirement (*please make all payoffs integers between 0 and 9, and underline the best responses in the payoff matrix*), or (ii) explain why the requirement is impossible to satisfy. Do not choose examples of games covered in class, discussion, or in the textbook.
- One player receives the same payoff regardless of the chosen strategies and neither player has a dominant strategy. (i.e., one player will receive the same payoff no matter what strategies they choose)
  - There are exactly 3 pure-strategy Nash equilibria.
  - One player has a strictly dominant strategy and there are no pure-strategy Nash equilibria.
  - One player has a dominant strategy and there are exactly two pure-strategy Nash equilibria.
  - One player has a strictly dominant strategy, the second player has a dominant strategy, and there are four pure-strategy Nash equilibria.

4. (15 points) Consider each of the following statements. Is this statement correct or incorrect? If you think it is correct, give a brief (1-3 sentence) explanation for why. If you think it is incorrect, give a counterexample of a game, together with a brief (1-3 sentence) explanation.
- If a pair of strategies gives both players higher payoffs than a pure-strategy Nash equilibrium, then it cannot be a Nash equilibrium.
  - If a two-player game has a pure-strategy Nash equilibrium, then at least one player must be playing a dominant strategy in that equilibrium.
  - If  $(s_A, s_B)$  is a pure-strategy Nash equilibrium and  $s_A$  is Player A's unique best response to  $s_B$ , then any other pure-strategy Nash equilibrium must have Player B choose a strategy other than  $s_B$ .

5. (22 points) Consider the following game. Assume  $0 \leq x \leq 6$  and  $0 \leq y \leq 8$ .

| Player 1 / Player 2 | L    | C      | R              |
|---------------------|------|--------|----------------|
| U                   | 6, 8 | 2, 2   | 1, $x + y$     |
| M                   | 3, 4 | $x, y$ | 4, 3           |
| D                   | 2, 1 | 3, 4   | 7, $y - x + 2$ |

- Suppose  $x = 4$  and  $y = 5$ . Find all pure-strategy Nash equilibria.
- Now allow  $x$  and  $y$  to vary. Find the range of  $x$  and the range of  $y$  for which (M,C) is a pure-strategy Nash equilibrium.
- Suppose Player 2 is exactly indifferent between L and R when Player 1 chooses U, and is exactly indifferent between C and R when Player 1 chooses D. Find  $x$  and  $y$ . At these values of  $x$  and  $y$ , find all pure-strategy Nash equilibria.